LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 4. Operators and Expressions

Lecture: 33. Challenge-Quadratic Equations

Section 1: Lecture Summary

This lecture introduces the concept of quadratic equations and how to find their roots, focusing especially on implementing the root-finding formula using Python. It starts by defining what a quadratic equation is—an equation of degree two typically expressed in the form $ax^2 + bx + c = 0$, explaining the components such as coefficients a, b, and c. The idea of roots is clarified: values of x that satisfy the equation and make it equal zero. The lecture demonstrates that a quadratic equation has two roots and shows, via example, how substituting these root values yields zero. Then, the well-known quadratic formula for roots is presented:

$$
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
$$

The discriminant $b^2 - 4ac$ is explained as vital to determine the nature of roots, with a note that it must be non-negative for real roots. Next, the formula is worked through step-by-step using a specific example with $a=1$, $b=-5$, and $c=6$, confirming the roots from the earlier substitution example. The lecture then transitions into translating the mathematical formula into Python code, particularly using the `math.sqrt()` function to compute the square root. The correct formatting and bracketing of the formula in Python is emphasized to avoid errors. Finally, a complete Python program is demonstrated, which calculates and prints the two roots for given coefficients, encouraging students to try writing and testing the program with different values.

Section 2: Key Concepts and Explanations

- **Quadratic Equation**
  - An equation of degree 2 in the form:

$$
ax^2 + bx + c = 0
$$

  - a, b, and c are coefficients; $a \neq 0$.
  - Called quadratic because the highest power of x is 2.
  - Distinction between equation (has '= 0') and expression (no '=').

- **Roots of a Quadratic Equation**
  - Values of x which satisfy the equation, making it zero.
  - A quadratic equation always has two roots (may be real or complex).
  - Example: For $x^2 - 5x + 6 = 0$, roots are $x=2$ and $x=3$ since substituting these values results in zero.

- **Quadratic Formula for Roots**
  - Formula:

$$
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
$$

  - The "$\pm$" indicates two roots: one with plus, one with minus.
  - The expression under the square root is called the **discriminant**:

$$
D = b^2 - 4ac
$$

  - For **real roots**, $D \geq 0$.

- Calculation steps:

  - Compute discriminant \(D\).

  - Find square root of \(D\).

  - Calculate roots using plus and minus versions.

- **Implementing the Formula in Python**

  - Use the `math` module for square root: `math.sqrt()`.

  - Careful use of parentheses to ensure correct order of operations.

  - Express formula separately for root1 and root2 using "+" and "-" accordingly.

  - Example coefficients can be hardcoded or taken as inputs.

Section 3: Example Code and Use Cases

Coefficients of the quadratic equation ax^2 + bx + c = 0

Calculate discriminant

Calculate the two roots using the quadratic formula

```
import math

a = 1
b = -5
c = 6

d = b**2 - 4*a*c

root1 = (-b + math.sqrt(d)) / (2 * a)
root2 = (-b - math.sqrt(d)) / (2 * a)

print("Roots are:", root1, root2)
```

Explanation:

This program calculates and prints the roots of the quadratic equation \(x^2 - 5x + 6 = 0\). It imports the `math` module to use the `sqrt()` function to calculate the square root of the discriminant. The discriminant d is computed first to ensure that the square root applies to the correct value. Two roots are calculated separately by considering the plus and minus signs in the formula and printed out. The expected roots are 3 and 2, matching the manual calculation.

Section 4: Key Takeaways

- A quadratic equation is a polynomial equation of degree 2, generally written as $ax^2 + bx + c = 0$.

- Roots of the quadratic equation are values of x that make the equation equal zero.

- Every quadratic equation has two roots, which can be found using the quadratic formula:

$$
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
$$

- The discriminant $b^2 - 4ac$ determines the nature of the roots; it must be non-negative for real roots.

- In Python, `math.sqrt()` is used for square roots, and careful parentheses placement is critical to correctly apply the quadratic formula.

- The formula yields two roots corresponding to the plus and minus in the formula, calculated separately.

- Practicing with different values of a, b, and c helps in understanding and verifying the solution.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.

Finding Roots of Quadratic Equation

(Challenge)

Finding Roots of Quadratic Equation
(Challenge)

Finding Roots of Quadratic Equation
(Challenge)

What is a Quadratic Equation ?

Finding Roots of Quadratic Equation

(Challenge)

What is a Quadratic Equation ?

Roots of Quadratic Equation

Finding Roots of Quadratic Equation

(Challenge)

What is a Quadratic Equation ?

Roots of Quadratic Equation

Formula for Roots

Quadratic Equation

Quadratic Equation

$$ax^2 + bx + c = 0$$

Quadratic Equation

$$ax^2 + bx + c = 0$$

Quadratic Equation

$$ax^2 + bx + c = 0$$

Quadratic Equation

Degree

$$ax^{2} + bx + c = 0$$

Quadratic Equation

$$\mathbf{a}x^{2} + \mathbf{b}x + \mathbf{c} = 0$$

Quadratic Equation

Coefficient

$$ax^2 + bx + c = 0$$

Roots of Quadratic Equation

Roots of Quadratic Equation

$$a x^2 + bx + c = 0$$

Roots of Quadratic Equation

$$a X^2 + bX + c = 0$$

Roots of Quadratic Equation

$$a X^2 + bX + c = 0$$

$X = ?$

Roots of Quadratic Equation

$$1X^2 - 5X + 6 = 0$$

Roots of Quadratic Equation

$$X^2 - 5X + 6 = 0$$

Roots of Quadratic Equation

$$1X^2 - 5X + 6 = 0$$

$x = 2$

Roots of Quadratic Equation

$$1(2)^2 - 5(2) + 6 = 0$$

$x = 2$

Roots of Quadratic Equation

$$4 \quad - 5(2) + 6 = 0$$

$x = 2$

Roots of Quadratic Equation

$$4 \quad - 10 + 6 = 0$$

$x = 2$

Roots of Quadratic Equation

$$- 6 + 6 = 0$$

$x = 2$

Roots of Quadratic Equation

$$- \ 6 \ + 6 = 0$$

root

$x = 2$

Roots of Quadratic Equation

$$1X^2 - 5X + 6 = 0$$

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$1(3)^2 - 5(3) + 6 = 0$$

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$9 \quad - 5(3) + 6 = 0$$

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$9 \quad - 15 + 6 = 0$$

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$- \; 6 \; + 6 = 0$$

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$-\ 6\ + 6 = 0$$

Roots

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$1\,X^2 - 5X + 6 = 0$$

Roots

$x = 2$

$x = 3$

Roots of Quadratic Equation

$$1X^2 - 5X + 6 = 0$$

Roots

$x_1 = 2$

$x_2 = 3$

Roots of Quadratic Equation

Formula for Roots

Formula for Roots

Formula for Roots

$$a\,x^2 + bx + c = 0$$

Formula for Roots

$$a x^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots

$$a x^2 + bx + c = 0$$

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots

$$a\,x^2 + bx + c = 0$$

$$x = \frac{-\,b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

$b^2 - 4ac > = 0$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

$a = 1$

$b = -5$

$c = 6$

$$d = b^2 - 4ac$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$d = b^2 - 4ac$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$d = (-5)^2 - 4ac$

Formula for Roots

$$\mathbf{1}x^2 - \mathbf{5}x + \mathbf{6} = 0$$

a = 1

b = -5

c = 6

d = 25 – 4.1.6

Formula for Roots

$$\mathbf{1}x^2 - \mathbf{5}x + \mathbf{6} = 0$$

a = 1

b = -5

c = 6

d = 25 – 24 = 1

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

$a = 1$

$b = -5$

$c = 6$

$d = 25 - 24 = 1$

$$x = \frac{-b \pm \sqrt{1}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

$a = 1$

$b = -5$

$c = 6$

$$x = \frac{-b \pm \sqrt{1}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

$a = 1$

$b = -5$

$c = 6$

$$x = \frac{5 \pm \sqrt{1}}{2a}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

$a = 1$

$b = -5$

$c = 6$

$$x = \frac{5 \pm \sqrt{1}}{2.1}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$$x = \frac{5 \pm 1}{2}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

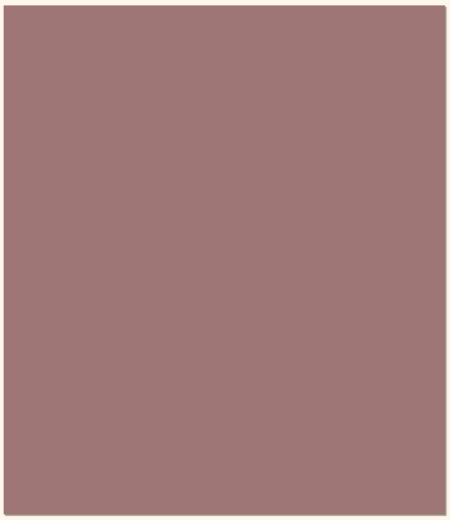
a = 1

b = -5

c = 6

$$x_1 = \frac{5 + 1}{2}$$

$$x_2 = \frac{5 - 1}{2}$$



Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$$x_1 = 3$$

$$x_2 = \frac{4}{2}$$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

a = 1

b = -5

c = 6

$x_1 = 3$

$x_2 = 2$

Formula for Roots

$$1x^2 - 5x + 6 = 0$$

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c = 6

$x_1 = 3$

$x_2 = 2$

Formula for Roots in Python

Formula for Roots in Python

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x = -b$

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x = -b \pm$

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x = -b \pm$ math.sqrt( )

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x = - b ± math.sqrt(b**2 – 4*a*c)

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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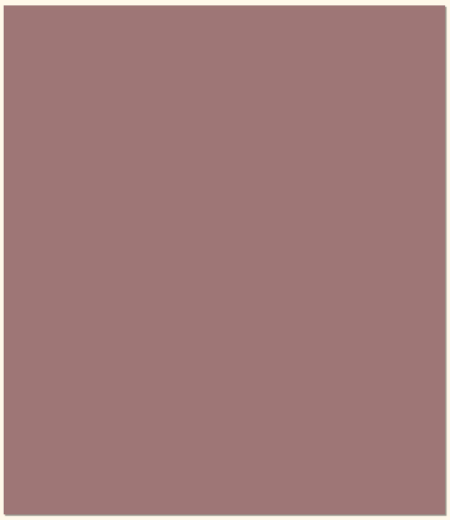

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =(- b ± math.sqrt(b**2 – 4*a*c))/(2*a)

Formula for Roots in Python

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x_1=(- b + math.sqrt(b**2 – 4*a*c))/**(2*a)**

x_2=(- b - math.sqrt(b**2 – 4*a*c))/**(2*a)**

Python Program

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```
import math
```

Python Program

```python
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c = 6
```

Python Program

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a = 1
b = -5
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r1 = (-b + math.sqrt(b**2 - 4*a*c))/(2*a)
r2 = (-b - math.sqrt(b**2 - 4*a*c))/(2*a)
```

Python Program

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r2 = (-b - math.sqrt(b**2 - 4*a*c))/(2*a)
print('Roots are ', r1, r2)
```

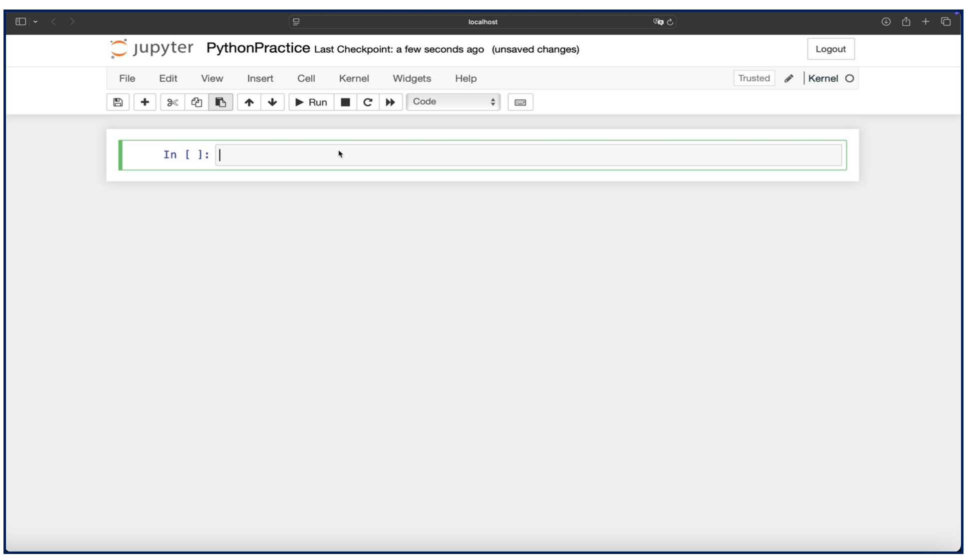

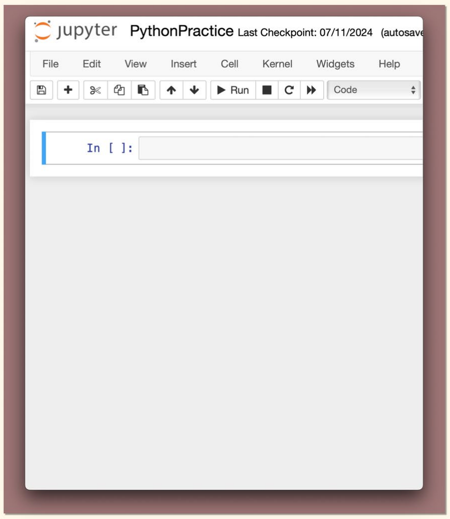

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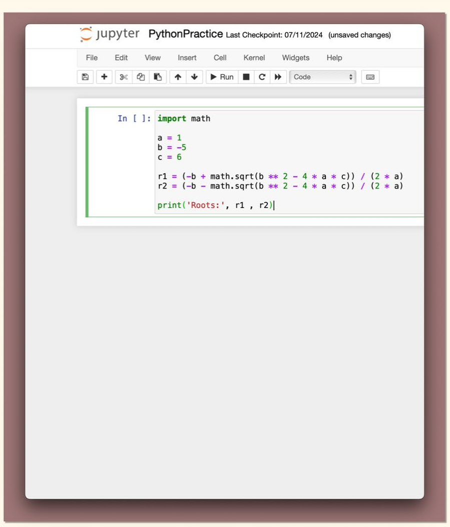

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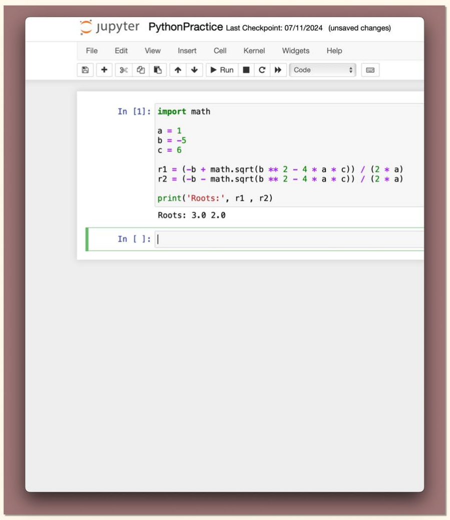

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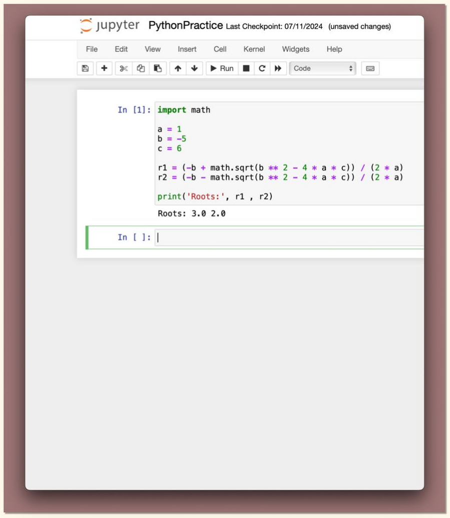
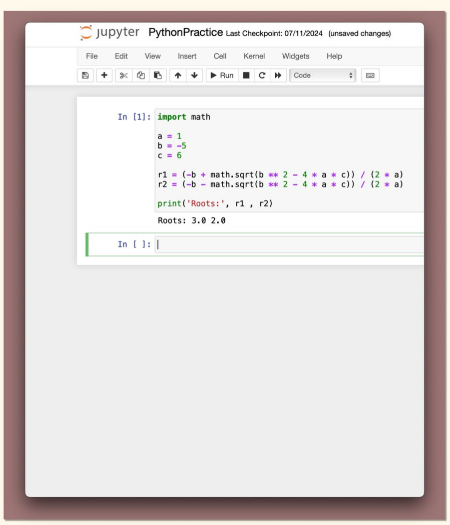


Python Program

```
import math
a = int(input('Enter a:'))
b = -5
c = 6
r1 = (-b + math.sqrt(b**2 - 4*a*c))/(2*a)
r2 = (-b - math.sqrt(b**2 - 4*a*c))/(2*a)
print('Roots are ', r1, r2)
```



Python Program

```
import math
a = int(input('Enter a:'))
b = int(input('Enter b:'))
c = 6
r1 = (-b + math.sqrt(b**2 - 4*a*c))/(2*a)
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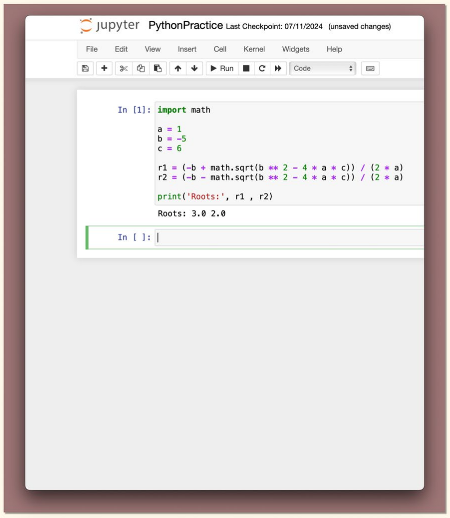

Python Program

```
import math
a = int(input('Enter a:'))
b = int(input('Enter b:'))
c = int(input('Enter c:'))
r1 = (-b + math.sqrt(b**2 - 4*a*c))/(2*a)
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